AI in customer service: triage now, auto-replies later

Solution brief for a retail bank ▪ Albin Oruglica ▪ September 2026 ▪ Based on a working prototype and a public dataset of 13,083 banking customer questions

The situation

A bank's service team answers thousands of short, often incomplete messages: *"my card hasn't arrived"*, *"why was I charged twice?"*. Answering them needs information from several systems. An AI assistant could handle many of them instantly, but a confident wrong answer to a customer is costly, and some cases (suspected fraud, disputes, anti-money-laundering checks) must never be decided by a machine. The business question is **which messages AI should handle, and at what point it should bring in a human.**

What was built and tested

A prototype that reads each message, identifies the request, looks up the customer's (simulated) card, account and transaction data, and then answers, asks one clarifying question, or hands over to an agent with a case summary. It was tested on a public dataset of banking customer questions, with fictional customer data. Two AI components were compared: a small, cheap **classifier** that sorts messages, and a **large language model** (Claude) that looks up data and writes replies.

Key findings

Sorting works.
The classifier identified the customer's request correctly in **89%** of cases, at practically zero cost, and knows when it is unsure.

Safety is a setting.
At a strict setting, **none of 440** fraud, dispute and money-laundering cases slipped past to automation; a looser setting automates more but lets about 1% through.

Replies are not ready.
Even with the request correctly identified, the AI fully resolved only **1 in 5** messages, and **18%** of its answers contained an error.

- **Where the value is.** Sorting messages and routing them to the right place could be worth up to about **€2.2–2.6 per message** against today's fully manual handling (assumed €5 per message). This is a ceiling: it assumes every sorted message then gets a correct reply.
- **Why not let the AI reply on its own.** When the AI's own replies were checked end to end, the fully automated system cost about the same as, or slightly more than, human-only handling (€5.1–6.0 vs. €5.00 per message). The errors were plausible-sounding: a real rule applied to the wrong product, an invented policy. Simple automatic checks do not catch them.
- **The right threshold depends on the bank's own costs** (what a wrong answer and an agent's time really cost, and how much risk is acceptable), and it must be re-measured whenever the model changes. It cannot be set once from a formula.

Recommendation

Phase 1: now — **AI triage and drafting, humans send.** The classifier sorts every message and routes fraud, dispute and anti-money-laundering cases straight to specialists. For the rest, the AI prepares a draft reply and a case summary, and the agent reviews, edits and sends. Low risk; no AI text reaches a customer unchecked.

Phase 2: later — **Auto-send for a narrow set of simple requests**, only once measured reply accuracy on the bank's own data beats the break-even (roughly fewer than 1 wrong answer in 6 at these costs), backed by a proper knowledge base of products and procedures, and with the strict safety setting.

Controls from day one: fixed, auditable rules for cases AI must never decide; monitoring of how many sensitive cases the sorting lets through; the AI sees only the customer data it needs; a data-processing agreement and legal basis for any external AI service (GDPR), or an in-house model if data must stay internal.

Proposed next step: a 6-week pilot with one service team

Run Phase 1 on live traffic and measure what the prototype could not: **agent handling time with and without AI drafts**, the share of drafts sent unchanged, sorting accuracy on the bank's real mix of requests, and the number of sensitive cases misrouted. These four numbers decide whether Phase 1 pays for itself and when Phase 2 is justified.

Evidence: prototype, code and all results in the project repository (see its README; every number can be reproduced with one command). Tested on the public BANKING77 dataset (PolyAI, CC BY 4.0) with invented customer data and an 11-line fictional product fact sheet; results are indicative, not a forecast for a specific bank. Cost figures use assumptions (€5 per manual message, €30 per wrong answer, €300 per misrouted sensitive case) and are tested for sensitivity in the repository.